

## 부속서 D 제품 환경성 정보(환경성적) 산정

### D.1 제품 환경성 정보(환경성적) 산정

제품 환경성 정보(환경성적) 산정은 전과정 목록분석(LCI) 결과를 특정 환경 영향범주 및 범주지표들과 연관시켜 잠재적 환경 영향의 중요성을 평가하는 것으로, LCA 연구에서의 영향평가(LCIA) 단계와 같다. 제품 환경성 정보(환경성적) 산정은 그림 D.1 에 따라 수행한다.

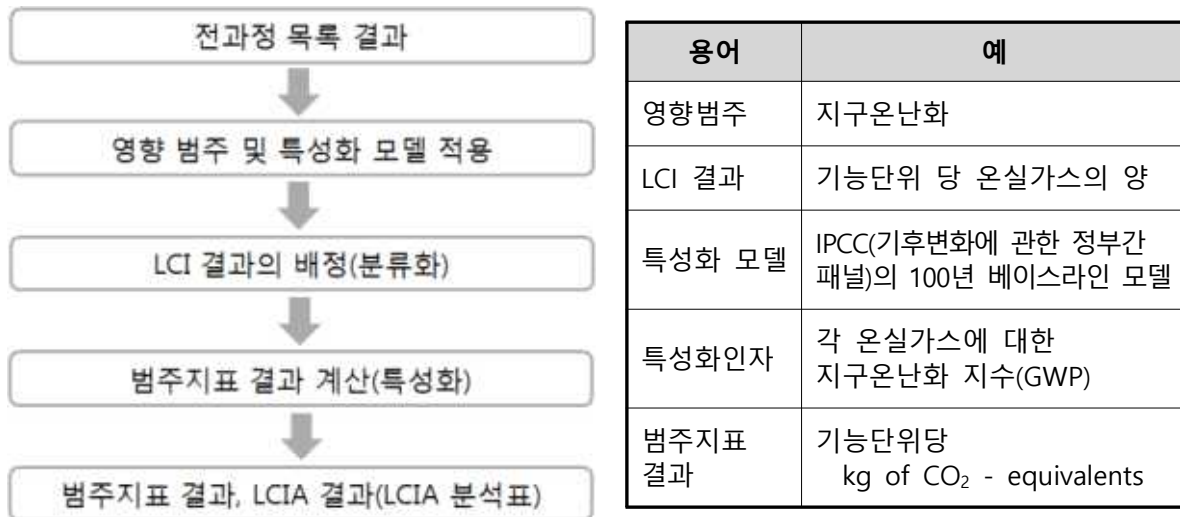


그림 D.1 — 제품 환경성 정보(환경성적) 산정

그림 D.2 — 용어의 예

### D.2 영향범주별 특성화 인자

#### D.2.1 자원소모(자원발자국)

출처 : Guinee et al. 2001 with modification for crude oil, natural gas, hard coal and soft coal. Reserve basis

| 구분 | 목록항목          | CAS No.   | ADP <sup>주1)</sup><br>(kg Sb-eq.) |
|----|---------------|-----------|-----------------------------------|
| 자원 | Actinium(Ac)  | 7440-34-8 | 6.33E+13                          |
|    | Aluminium(Al) | 7429-90-5 | 1.00E-08                          |
|    | Antimony(Sb)  | 7440-36-0 | 1.00E+00                          |
|    | Argon(Ar)     | 7440-37-1 | 4.71E-07                          |
|    | Arsenic(As)   | 7440-38-2 | 9.17E-03                          |
|    | Barium(Ba)    | 7440-39-3 | 1.06E-10                          |
|    | Beryllium(Be) | 7440-41-7 | 3.19E-05                          |
|    | Bismuth(Bi)   | 7440-69-9 | 7.31E-02                          |
|    | Boron(B)      | 7440-42-8 | 4.67E-03                          |
|    | Bromine(Br)   | 7726-95-6 | 6.67E-03                          |
|    | Cadmium(Cd)   | 7440-43-9 | 3.30E-01                          |
|    | Calcium(Ca)   | 7440-70-2 | 7.08E-10                          |

| 구분 | 목록항목           | CAS No.   | ADP <sup>註1)</sup><br>(kg Sb-eq.) |
|----|----------------|-----------|-----------------------------------|
|    | Cerium(Ce)     | 7440-45-1 | 5.32E-09                          |
|    | Cesium(Cs)     | 7440-46-2 | 1.91E-05                          |
|    | Chlorine(Cl)   | 7782-50-5 | 4.86E-08                          |
|    | Chromium(Cr)   | 7440-47-3 | 8.58E-04                          |
|    | Cobalt(Co)     | 7440-48-4 | 2.62E-05                          |
|    | Copper(Cu)     | 7440-50-8 | 1.94E-03                          |
|    | Dysprosium(Dy) | 7429-91-6 | 2.13E-06                          |
|    | Erbium(Er)     | 7440-52-0 | 2.44E-06                          |
|    | Europium(Eu)   | 7440-53-1 | 1.33E-05                          |
|    | Fluorine(F)    | 7782-41-4 | 2.96E-06                          |
|    | Gadolinium(Gd) | 7440-54-2 | 6.57E-07                          |
|    | Gallium(Ga)    | 7440-55-3 | 1.03E-07                          |
|    | Germanium(Ge)  | 7440-56-4 | 1.47E-06                          |
|    | Gold(Au)       | 7440-57-5 | 8.95E+01                          |
|    | Hafnium(Hf)    | 7440-58-6 | 8.67E-07                          |
|    | Helium(He)     | 7440-59-7 | 1.48E+02                          |
|    | Holmium(Ho)    | 7440-60-0 | 1.33E-05                          |
|    | Indium(In)     | 7440-74-6 | 9.03E-03                          |
|    | Iodine(I)      | 7553-56-2 | 4.27E-02                          |
|    | Iridium(Ir)    | 7439-88-5 | 3.23E+01                          |
|    | Iron(Fe)       | 7439-89-6 | 8.43E-08                          |
|    | Kalium(K)      | 7440-09-7 | 3.13E-08                          |
|    | Krypton(Kr)    | 7439-90-9 | 2.09E+01                          |
|    | Lanthanum(La)  | 7439-91-0 | 2.13E-08                          |
|    | Lead(Pb)       | 7439-92-1 | 1.35E-02                          |
|    | Lithium(Li)    | 7439-93-2 | 9.23E-06                          |
|    | Lutetium(Lu)   | 7439-94-3 | 7.66E-05                          |
|    | Magnesium(Mg)  | 7439-95-4 | 3.73E-09                          |
|    | Manganese(Mn)  | 7439-96-5 | 1.38E-05                          |
|    | Mercury(Hg)    | 7439-97-6 | 4.95E-01                          |
|    | Molybdenum(Mo) | 7439-98-7 | 3.17E-02                          |
|    | Neodymium(Nd)  | 7440-00-8 | 1.94E-17                          |
|    | Neon(Ne)       | 7440-01-9 | 3.25E-01                          |
|    | Nickel(Ni)     | 7440-02-0 | 1.08E-04                          |
|    | Niobium(Nb)    | 7440-03-1 | 2.31E-05                          |
|    | Osmium(Os)     | 7440-04-2 | 1.44E+01                          |
|    | Palladium(Pd)  | 7440-05-3 | 3.23E-01                          |
|    | Phosphorus(P)  | 7723-14-0 | 8.44E-05                          |

| 구분 | 목록항목                       | CAS No.    | ADP <sup>주1)</sup><br>(kg Sb-eq.) |
|----|----------------------------|------------|-----------------------------------|
|    | Platinum(Pt)               | 7440-06-4  | 1.29E+00                          |
|    | Polonium(Po)               | 7440-08-6  | 4.79E+14                          |
|    | Praseodymium(Pr)           | 7440-10-0  | 2.85E-07                          |
|    | Protactinium(Pa)           | 7440-13-3  | 9.77E+06                          |
|    | Radium(Ra)                 | 7440-14-4  | 2.36E+07                          |
|    | Radon(Rn)                  | 10043-92-2 | 1.20E+20                          |
|    | Rhenium(Re)                | 7440-15-5  | 7.66E-01                          |
|    | Rhodium(Rh)                | 7440-16-6  | 3.23E+01                          |
|    | Rubidium(Rb)               | 7440-17-7  | 2.36E-09                          |
|    | Ruthenium(Ru)              | 7440-18-8  | 3.23E+01                          |
|    | Samarium(Sm)               | 7440-19-9  | 5.32E-07                          |
|    | Scandium(Sc)               | 7440-20-2  | 3.96E-08                          |
|    | Selenium(Se)               | 7782-49-2  | 4.75E-01                          |
|    | Silicium(Si)               | 7440-21-3  | 2.99E-11                          |
|    | Silver(Ag)                 | 7440-22-4  | 1.84E+00                          |
|    | Sodium(Na)                 | 7440-23-5  | 8.24E-11                          |
|    | Strontium(Sr)              | 7440-24-6  | 1.12E-06                          |
|    | Sulfur(S)                  | 7704-34-9  | 3.58E-04                          |
|    | Tantalum(Ta)               | 7440-25-7  | 6.77E-05                          |
|    | Tellurium(Te)              | 13494-80-9 | 5.28E+01                          |
|    | Terbium(Tb)                | 7440-27-9  | 2.36E-05                          |
|    | Thallium(Tl)               | 7440-28-0  | 5.05E-05                          |
|    | Thorium(Th)                | 7440-29-1  | 2.08E-07                          |
|    | Thulium(Tm)                | 7440-30-4  | 8.31E-05                          |
|    | Tin(Sn)                    | 7440-31-5  | 3.30E-02                          |
|    | Titanium(Ti)               | 7440-32-6  | 4.40E-08                          |
|    | Tungsten(W)                | 7440-33-7  | 1.17E-02                          |
|    | Uranium(U)                 | 7440-61-1  | 2.87E-03                          |
|    | Vanadium(V)                | 7440-62-2  | 1.16E-06                          |
|    | Xenon(Xe)                  | 7440-63-3  | 1.75E+04                          |
|    | Ytterbium(Yb)              | 7440-64-4  | 2.13E-06                          |
|    | Yttrium(Y)                 | 7440-65-5  | 3.34E-07                          |
|    | Zinc(Zn)                   | 7440-66-6  | 9.92E-04                          |
|    | Zirconium(Zr)              | 7440-67-7  | 1.86E-05                          |
|    | Hard coal                  |            | 1.34E-02                          |
|    | Soft coal                  |            | 6.71E-03                          |
|    | Natural gas <sup>주2)</sup> | 8006-14-2  | 1.51E-02                          |

| 구분 | 목록항목                               | CAS No.      | ADP <sup>주1)</sup><br>(kg Sb-eq.) |
|----|------------------------------------|--------------|-----------------------------------|
|    | Crude oil                          | 8012-95-1    | 2.01E-02                          |
|    | Bauxite( $\text{Al}_2\text{O}_3$ ) | 7429-90-5(o) | 2.10E-09                          |
|    | fossil energy                      | -            | 4.81E-04                          |

주1) ADP : Abiotic Depletion Potential

주2) Natural gas : 1.87E-02 m<sup>3</sup>에 0.806 kg/Nm<sup>3</sup> 환산인자 적용한 값

## D.2.2 지구온난화(탄소발자국)

- 출처 : 기후변화에 관한 정부간 패널(IPCC)의 1996년 보고 지침에 의해 발행된 100-year time horizon에 대한 다양한 지구온난화 지수

| 구분 | 목록항목                                                         | CAS No.     | GWP <sup>주)</sup><br>(kg CO <sub>2</sub> -eq.) |
|----|--------------------------------------------------------------|-------------|------------------------------------------------|
| 대기 | Carbon dioxide(CO <sub>2</sub> )                             | 124-38-9    | 1.00E+00                                       |
|    | Methane(CH <sub>4</sub> )                                    | 74-82-8     | 2.10E+01                                       |
|    | Nitrous oxide(N <sub>2</sub> O)                              | 10024-97-2  | 3.10E+02                                       |
|    | HFC-23(CHF <sub>3</sub> )                                    | 75-46-7     | 1.17E+04                                       |
|    | HFC-32(CH <sub>2</sub> F <sub>2</sub> )                      | 75-10-5     | 6.50E+02                                       |
|    | HFC-41(CH <sub>3</sub> F)                                    | 593-53-3    | 1.50E+02                                       |
|    | HFC-43-10mee(C <sub>5</sub> H <sub>2</sub> F <sub>10</sub> ) | 138495-42-8 | 1.30E+03                                       |
|    | HFC-125(C <sub>2</sub> HF <sub>5</sub> )                     | 354-33-6    | 2.80E+03                                       |
|    | HFC-134(C <sub>2</sub> H <sub>2</sub> F <sub>4</sub> )       | 811-97-2    | 1.00E+03                                       |
|    | HFC-134a(CH <sub>2</sub> FCF <sub>3</sub> )                  | 811-97-2(a) | 1.30E+03                                       |
|    | HFC-152a(C <sub>2</sub> H <sub>4</sub> F <sub>2</sub> )      | 75-37-6     | 1.40E+02                                       |
|    | HFC-143(C <sub>2</sub> H <sub>3</sub> F <sub>3</sub> )       | 430-66-0    | 3.00E+02                                       |
|    | HFC-143a(C <sub>2</sub> H <sub>3</sub> F <sub>3</sub> )      | 420-46-2    | 3.80E+03                                       |
|    | HFC-227ea(C <sub>3</sub> HF <sub>7</sub> )                   | 431-89-0    | 2.90E+03                                       |
|    | HFC-236fa(C <sub>3</sub> H <sub>2</sub> F <sub>6</sub> )     | 690-39-1    | 6.30E+03                                       |
|    | HFC-245ca(C <sub>3</sub> H <sub>3</sub> F <sub>5</sub> )     | 1814-88-6   | 5.60E+02                                       |
|    | Sulphur hexafluoride(SF <sub>6</sub> )                       | 2551-62-4   | 2.39E+04                                       |
|    | Perfluoromethane(CF <sub>4</sub> )                           | 75-73-0     | 6.50E+03                                       |
|    | Perfluoroethane(C <sub>2</sub> F <sub>6</sub> )              | 76-16-4     | 9.20E+03                                       |
|    | Perfluoropropane(C <sub>3</sub> F <sub>8</sub> )             | 76-19-7     | 7.00E+03                                       |
|    | Perfluorobutane(C <sub>4</sub> F <sub>10</sub> )             | 355-25-9    | 7.00E+03                                       |
|    | Perfluorocyclobutane(c-C <sub>4</sub> F <sub>8</sub> )       | 115-25-3    | 8.70E+03                                       |
|    | Perfluoropentane(C <sub>5</sub> F <sub>12</sub> )            | 678-26-2    | 7.50E+03                                       |
|    | Perfluorohexane(C <sub>6</sub> F <sub>14</sub> )             | 355-42-0    | 7.40E+03                                       |

주) GWP : Global Warming Potential

### D.2.3 오존층영향

- 출처 : WMO(World Metrological Organizaion), 1999

| 구분 | 목록항목                  | CAS No.        | ODP <sup>주)</sup><br>(kg CFC-11-eq.) |
|----|-----------------------|----------------|--------------------------------------|
| 대기 | 1,1,1-Trichloroethane | 71-55-6        | 1.10E-01                             |
|    | CFC-11                | 75-69-4        | 1.00E+00                             |
|    | CFC-113               | 76-13-1        | 9.00E-01                             |
|    | CFC-114               | 76-14-2        | 8.50E-01                             |
|    | CFC-115               | 76-15-3        | 4.00E-01                             |
|    | CFC-12                | 75-71-8        | 8.20E-01                             |
|    | HALON-1202            | 75-61-6        | 1.25E+00                             |
|    | HALON-1211            | 353-59-3       | 5.10E+00                             |
|    | HALON-1301            | 75-63-8        | 1.20E+01                             |
|    | HALON-2402            | 124-73-2       | 7.00E+00                             |
|    | HBFC-1201             | 1511-62-2      | 1.40E+00                             |
|    | HBFC-2311             | 151-67-7       | 1.40E-01                             |
|    | HBFC-2401             | 124-72-1 (bis) | 2.50E-01                             |
|    | HCFC-123              | 306-83-2       | 1.20E-02                             |
|    | HCFC-124              | 2837-89-0      | 2.60E-02                             |
|    | HCFC-141b             | 1717-00-6      | 8.60E-02                             |
|    | HCFC-142b             | 75-68-3        | 4.30E-02                             |
|    | HCFC-22               | 75-45-6        | 3.40E-02                             |
|    | HCFC-225ca            | 422-56-0       | 1.70E-02                             |
|    | HCFC-225cb            | 507-55-1       | 1.70E-02                             |
|    | Methyl chloride       | 74-87-3        | 2.00E-02                             |
|    | Methyl bromide        | 74-83-9        | 3.70E-01                             |
|    | Tetrachloromethane    | 56-23-5        | 1.20E+00                             |

주) ODP : Ozone Depletion Potential

## D.2.4 산성화(산성비)

- 출처 : Heijungs et al, 1992 (updated with Hauschild & Wenzel, 1998)

| 구분 | 목록항목                                             | CAS No.    | AP <sup>주)</sup><br>(kg SO <sub>2</sub> -eq.) |
|----|--------------------------------------------------|------------|-----------------------------------------------|
| 대기 | Ammonia(NH <sub>3</sub> )                        | 7664-41-7  | 1.88E+00                                      |
|    | Hydrogen chloride(HCl)                           | 7647-01-0  | 8.80E-01                                      |
|    | Hydrogen fluoride(HF)                            | 7664-39-3  | 1.60E+00                                      |
|    | Hydrogen sulfide(H <sub>2</sub> S)               | 7783-06-4  | 1.88E+00                                      |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 5.10E-01                                      |
|    | Nitrogen dioxide(NO <sub>2</sub> )               | 10102-44-0 | 7.00E-01                                      |
|    | Nitrogen oxide(NO)                               | 10102-43-9 | 1.07E+00                                      |
|    | Nitrogen oxides(NO <sub>x</sub> )                | 10102-44-0 | 7.00E-01                                      |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.80E-01                                      |
|    | Sulfur dioxide(SO <sub>2</sub> )                 | 7446-09-5  | 1.00E+00                                      |
|    | Sulfur trioxide(SO <sub>3</sub> )                | 7446-11-9  | 8.00E-01                                      |
|    | Sulfuric acid(H <sub>2</sub> SO <sub>4</sub> )   | 7664-93-9  | 6.50E-01                                      |
| 수계 | Hydrogen chloride(HCl)                           | 7647-01-0  | 8.80E-01                                      |
|    | Hydrogen fluoride(HF)                            | 7664-39-3  | 1.60E+00                                      |
|    | Hydrogen sulfide(H <sub>2</sub> S)               | 7783-06-4  | 1.88E+00                                      |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 5.10E-01                                      |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.80E-01                                      |
|    | Sulfuric acid(H <sub>2</sub> SO <sub>4</sub> )   | 7664-93-9  | 6.50E-01                                      |
| 토양 | Hydrogen chloride(HCl)                           | 7647-01-0  | 8.80E-01                                      |
|    | Hydrogen fluoride(HF)                            | 7664-39-3  | 1.60E+00                                      |
|    | Hydrogen sulfide(H <sub>2</sub> S)               | 7783-06-4  | 1.88E+00                                      |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 5.10E-01                                      |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.80E-01                                      |
|    | Sulfuric acid(H <sub>2</sub> SO <sub>4</sub> )   | 7664-93-9  | 6.50E-01                                      |

주) AP : Acidification Potential

## D.2.5 부영양화

- 출처 : Heijungs et al, 1992 (with some modifications)

| 구분 | 목록항목                                             | CAS No.    | EP <sup>주)</sup><br>(kg PO <sub>4</sub> <sup>3-</sup> -eq.) |
|----|--------------------------------------------------|------------|-------------------------------------------------------------|
| 대기 | Ammonia(NH <sub>3</sub> )                        | 7664-41-7  | 3.50E-01                                                    |
|    | Ammonium(NH <sub>4</sub> <sup>+</sup> )          | 14798-03-9 | 3.30E-01                                                    |
|    | Nitrate(NO <sub>3</sub> <sup>-</sup> )           | 14797-55-8 | 1.00E-01                                                    |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 1.00E-01                                                    |
|    | Nitrogen(N <sub>2</sub> )                        | 7727-37-9  | 4.20E-01                                                    |
|    | Nitrogen dioxide(NO <sub>2</sub> )               | 10102-44-0 | 1.30E-01                                                    |
|    | Nitrogen oxide(NO)                               | 10102-43-9 | 2.00E-01                                                    |
|    | Nitrogen oxides(NO <sub>x</sub> )                | 10102-44-0 | 1.30E-01                                                    |
|    | Phosphate(PO <sub>4</sub> <sup>3-</sup> )        | 14265-44-2 | 1.00E+00                                                    |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.70E-01                                                    |
|    | Phosphorus(P)                                    | 7723-14-0  | 3.06E+00                                                    |
|    | Phosphorus oxide(P <sub>2</sub> O <sub>5</sub> ) | 1314-56-3  | 1.34E+00                                                    |
| 수계 | Ammonia(NH <sub>3</sub> )                        | 7664-41-7  | 3.50E-01                                                    |
|    | Ammonium(NH <sub>4</sub> <sup>+</sup> )          | 14798-03-9 | 3.30E-01                                                    |
|    | COD                                              | -          | 2.20E-02                                                    |
|    | Nitrate(NO <sub>3</sub> <sup>-</sup> )           | 14797-55-8 | 1.00E-01                                                    |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 1.00E-01                                                    |
|    | Nitrite(NO <sub>2</sub> <sup>-</sup> )           | 14797-65-0 | 1.00E-01                                                    |
|    | Nitrogen(N <sub>2</sub> )                        | 7727-37-9  | 4.20E-01                                                    |
|    | Phosphate(PO <sub>4</sub> <sup>3-</sup> )        | 14265-44-2 | 1.00E+00                                                    |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.70E-01                                                    |
|    | Phosphorus(P)                                    | 7723-14-0  | 3.06E+00                                                    |
|    | Phosphorus oxide(P <sub>2</sub> O <sub>5</sub> ) | 1314-56-3  | 1.34E+00                                                    |
| 토양 | Ammonia(NH <sub>3</sub> )                        | 7664-41-7  | 3.50E-01                                                    |
|    | Ammonium(NH <sub>4</sub> <sup>+</sup> )          | 14798-03-9 | 3.30E-01                                                    |
|    | Nitrate(NO <sub>3</sub> <sup>-</sup> )           | 14797-55-8 | 1.00E-01                                                    |
|    | Nitric acid(HNO <sub>3</sub> )                   | 7697-37-2  | 1.00E-01                                                    |
|    | Nitrogen(N <sub>2</sub> )                        | 7727-37-9  | 4.20E-01                                                    |
|    | Phosphate(PO <sub>4</sub> <sup>3-</sup> )        | 14265-44-2 | 1.00E+00                                                    |
|    | Phosphoric acid(H <sub>3</sub> PO <sub>4</sub> ) | 7664-38-2  | 9.70E-01                                                    |
|    | Phosphorus(P)                                    | 7723-14-0  | 3.06E+00                                                    |
|    | Phosphorus oxide(P <sub>2</sub> O <sub>5</sub> ) | 1314-56-3  | 1.34E+00                                                    |

주) EP : Eutrophication Potential

## D.2.6 광화학적 산화물생성(광화학 스모그)

- 출처 : Jenkin & Hayman, 1999; Derwent et al. 1998; high NOx

| 구분 | 목록항목                     | CAS No.    | POCP <sup>주)</sup><br>(kg C <sub>2</sub> H <sub>4</sub> -eq.) |
|----|--------------------------|------------|---------------------------------------------------------------|
| 대기 | 1,1,1-Trichloroethane    | 71-55-6    | 9.00E-03                                                      |
|    | 1,2,3-Trimethylbenzene   | 526-73-8   | 1.27E+00                                                      |
|    | 1,2,4-Trimethylbenzene   | 95-63-6    | 1.28E+00                                                      |
|    | 1,3,5-Trimethylbenzene   | 108-67-8   | 1.38E+00                                                      |
|    | 1,3-Butadiene            | 106-99-0   | 8.50E-01                                                      |
|    | 1-Butanol                | 71-36-3    | 6.20E-01                                                      |
|    | But-1-ene                | 106-98-9   | 1.08E+00                                                      |
|    | 1-Butoxypropanol         | 57018-52-7 | 4.63E-01                                                      |
|    | 1-Butyl acetate          | 123-86-4   | 2.69E-01                                                      |
|    | Hex-1-ene                | 592-41-6   | 8.74E-01                                                      |
|    | 1-Methoxy-2-propanol     | 107-98-2   | 3.55E-01                                                      |
|    | Pent-1-ene               | 109-67-1   | 9.77E-01                                                      |
|    | n-Propanol               | 71-23-8    | 5.61E-01                                                      |
|    | 1-Propylbenzene          | 103-65-1   | 6.36E-01                                                      |
|    | 1-Propylacetate          | 109-60-4   | 2.82E-01                                                      |
|    | n-Undecane               | 1120-21-4  | 3.84E-01                                                      |
|    | 2,2-Dimethylbutane       | 75-83-2    | 2.41E-01                                                      |
|    | 2,3-Dimethylbutane       | 79-29-8    | 5.41E-01                                                      |
|    | Methylethylketone        | 78-93-3    | 3.73E-01                                                      |
|    | 2-Butoxyethanol          | 111-76-2   | 4.83E-01                                                      |
|    | 2-Ethoxyethanol          | 110-80-5   | 3.86E-01                                                      |
|    | 2-Methoxyethanol         | 109-86-4   | 3.07E-01                                                      |
|    | 2-Methylbut-1-ene        | 563-46-2   | 7.71E-01                                                      |
|    | 2-Methylbut-2-ene        | 513-35-9   | 8.42E-01                                                      |
|    | 2-Methylbutan-1-ol       | 137-32-6   | 4.89E-01                                                      |
|    | 2-Methylbutan-2-ol       | 75-85-4    | 2.28E-01                                                      |
|    | 2-Methylhexane           | 591-76-4   | 4.11E-01                                                      |
|    | 2-Methylpentane          | 107-83-5   | 4.20E-01                                                      |
|    | 3,5-Diethyltoluene       | 25550-13-4 | 1.30E+00                                                      |
|    | 3,5-Dimethylethylbenzene | 29224-55-3 | 1.32E+00                                                      |
|    | 3-Methylbut-1-ene        | 563-45-1   | 6.71E-01                                                      |
|    | 3-Methylbutan-1-ol       | 123-51-3   | 4.33E-01                                                      |
|    | 3-Methylbutan-2-ol       | 598-75-4   | 4.06E-01                                                      |
|    | 3-Methylhexane           | 589-34-4   | 3.64E-01                                                      |
|    | 3-Methylpentane          | 96-14-0    | 4.79E-01                                                      |
|    | 3-Pentanol               | 584-02-1   | 5.95E-01                                                      |

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|----|------------------------|-----------|---------------------------------------------------------------|
|    | Acetaldehyde           | 75-07-0   | 6.41E-01                                                      |
|    | Acetic acid            | 64-19-7   | 9.70E-02                                                      |
|    | Acetone                | 67-64-1   | 9.40E-02                                                      |
|    | Acetylene              | 74-86-2   | 8.50E-02                                                      |
|    | Benzaldehyde           | 100-52-7  | -9.20E-02                                                     |
|    | Benzene                | 71-43-2   | 2.18E-01                                                      |
|    | n-Butane               | 106-97-8  | 3.52E-01                                                      |
|    | Butyraldehyde          | 123-72-8  | 7.95E-01                                                      |
|    | Carbon monoxide(CO)    | 630-08-0  | 2.70E-02                                                      |
|    | cis-But-2-ene          | 590-18-1  | 1.15E+00                                                      |
|    | cis-Hex-2-ene          | 7688-21-3 | 1.07E+00                                                      |
|    | cis-Pent-2-ene         | 627-20-3  | 1.12E+00                                                      |
|    | cis-1,2-Dichloroethene | 156-59-2  | 4.47E-01                                                      |
|    | Cyclohexane            | 110-82-7  | 2.90E-01                                                      |
|    | Cyclohexanol           | 108-93-0  | 5.18E-01                                                      |
|    | Cyclohexanone          | 108-94-1  | 2.99E-01                                                      |
|    | n-Decane               | 124-18-5  | 3.84E-01                                                      |
|    | Diacetone alcohol      | 123-42-2  | 3.07E-01                                                      |
|    | Dichloromethane        | 75-09-2   | 6.80E-02                                                      |
|    | Diethylether           | 60-29-7   | 4.45E-01                                                      |
|    | Diethylketone          | 96-22-0   | 4.14E-01                                                      |
|    | Diisopropylether       | 108-20-3  | 3.98E-01                                                      |
|    | Dimethoxy methane      | 109-87-5  | 1.60E-01                                                      |
|    | Dimethyl carbonate     | 616-38-6  | 2.50E-02                                                      |
|    | Dimethylether          | 115-10-6  | 1.89E-01                                                      |
|    | n-Dodecane             | 112-40-3  | 3.57E-01                                                      |
|    | Ethane                 | 74-84-0   | 1.23E-01                                                      |
|    | Ethanol                | 64-17-5   | 3.99E-01                                                      |
|    | Ethyl acetate          | 141-78-6  | 2.09E-01                                                      |
|    | Ethyl-t-butylether     | 637-92-3  | 2.44E-01                                                      |
|    | Ethylbenzene           | 100-41-4  | 7.30E-01                                                      |
|    | Ethylene               | 74-85-1   | 1.00E+00                                                      |
|    | Ethylene glycol        | 107-21-1  | 3.73E-01                                                      |
|    | Formaldehyde           | 50-00-0   | 5.19E-01                                                      |
|    | Formic acid            | 64-18-6   | 3.20E-02                                                      |
|    | n-Heptane              | 142-82-5  | 4.94E-01                                                      |
|    | Hexan-2-one            | 591-78-6  | 5.72E-01                                                      |
|    | Hexan-3-one            | 589-38-8  | 5.99E-01                                                      |
|    | n-Hexane               | 110-54-3  | 4.82E-01                                                      |
|    | i-Butane               | 75-28-5   | 3.07E-01                                                      |

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|----|------------------------------------|------------|---------------------------------------------------------------|
|    | i-Butanol                          | 78-83-1    | 3.60E-01                                                      |
|    | Methylpropene                      | 115-11-7   | 6.27E-01                                                      |
|    | i-Butyraldehyde                    | 78-84-2    | 5.14E-01                                                      |
|    | i-Pentane                          | 78-78-4    | 4.05E-01                                                      |
|    | Isoprene                           | 78-79-5    | 1.09E+00                                                      |
|    | i-Propanol                         | 67-63-0    | 1.88E-01                                                      |
|    | i-Propyl acetate                   | 108-21-4   | 2.11E-01                                                      |
|    | i-Propylbenzene                    | 98-82-8    | 5.00E-01                                                      |
|    | m-Ethyltoluene                     | 620-14-4   | 1.02E+00                                                      |
|    | m-Xylene                           | 108-38-3   | 1.11E+00                                                      |
|    | Methane                            | 74-82-8    | 6.00E-03                                                      |
|    | Methanol                           | 67-56-1    | 1.40E-01                                                      |
|    | Methyl acetate                     | 79-20-9    | 5.90E-02                                                      |
|    | Methyl chloride                    | 74-87-3    | 5.00E-03                                                      |
|    | Methyl formate                     | 107-31-3   | 2.70E-02                                                      |
|    | Methyl-i-butylketone               | 108-10-1   | 4.90E-01                                                      |
|    | Methylpropylketone                 | 107-87-9   | 5.48E-01                                                      |
|    | Methyl-t-butylether                | 1634-04-4  | 1.75E-01                                                      |
|    | Methyl-t-butylketone               | 75-97-8    | 3.23E-01                                                      |
|    | Methyl-i-propylketone              | 563-80-4   | 3.64E-01                                                      |
|    | Neopentane                         | 463-82-1   | 1.73E-01                                                      |
|    | Nitrogen dioxide(NO <sub>2</sub> ) | 10102-44-0 | 2.80E-02                                                      |
|    | Nitrogen oxide(NO)                 | 10102-43-9 | -4.27E-01                                                     |
|    | n-Nonane                           | 111-84-2   | 4.14E-01                                                      |
|    | n-Octane                           | 111-65-9   | 4.53E-01                                                      |
|    | o-Ethyltoluene                     | 611-14-3   | 8.98E-01                                                      |
|    | o-Xylene                           | 95-47-6    | 1.05E+00                                                      |
|    | p-Ethyltoluene                     | 622-96-8   | 9.06E-01                                                      |
|    | p-Xylene                           | 106-42-3   | 1.01E+00                                                      |
|    | Pentanaldehyde                     | 110-62-3   | 7.65E-01                                                      |
|    | n-Pentane                          | 109-66-0   | 3.95E-01                                                      |
|    | Propane                            | 74-98-6    | 1.76E-01                                                      |
|    | Propionic acid                     | 79-09-4    | 1.50E-01                                                      |
|    | Propionaldehyde                    | 123-38-6   | 7.98E-01                                                      |
|    | Propene                            | 115-07-1   | 1.12E+00                                                      |
|    | Propylene glycol                   | 57-55-6    | 4.57E-01                                                      |
|    | s-Butanol                          | 78-92-2    | 4.00E-01                                                      |
|    | s-Butyl acetate                    | 105-46-4   | 2.75E-01                                                      |
|    | Styrene                            | 100-42-5   | 1.42E-01                                                      |
|    | Sulfur dioxide(SO <sub>2</sub> )   | 7446-09-5  | 4.80E-02                                                      |

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|----|------------------------|-----------|---------------------------------------------------------------|
|    | t-Butanol              | 75-65-0   | 1.06E-01                                                      |
|    | t-Butyl acetate        | 540-88-5  | 5.30E-02                                                      |
|    | Tetrachloroethylene    | 127-18-4  | 2.90E-02                                                      |
|    | Toluene                | 108-88-3  | 6.37E-01                                                      |
|    | trans-But-2-ene        | 624-64-6  | 1.13E+00                                                      |
|    | trans-Hex-2-ene        | 4050-45-7 | 1.07E+00                                                      |
|    | trans-Pent-2-ene       | 646-04-8  | 1.12E+00                                                      |
|    | trans-Dichloroethylene | 156-60-5  | 3.92E-01                                                      |
|    | Trichloroethylene      | 79-01-6   | 3.25E-01                                                      |
|    | Trichloromethane       | 67-66-3   | 2.30E-02                                                      |
|    | NMVOC                  | -         | 1.00E+00                                                      |

주) POCP : Photochemical Ozone Creation Potential